LIVE CLASSROOM ATTENDANCE COUNTER

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Abstract –*In educational institutions, student attendance is usually recorded manually, which is time-consuming, labor-intensive, and prone to errors such as proxy attendance. This traditional method reduces effective teaching time and increases the workload on faculty members. To address this problem, this paper proposes a Live Classroom Attendance System using Computer Vision. The system uses a webcam to capture real-time video of the classroom and automatically detect student movement. Background subtraction and object detection techniques are applied to identify students entering and exiting the classroom. A virtual reference line is used to count entries and exits accurately. The system continuously calculates and displays the number of students present inside the classroom. Audio alerts are also provided for entry and exit events. The proposed system improves accuracy, saves time, and minimizes human intervention. It provides a cost-effective and efficient solution for smart classroom environments. Future enhancements include face recognition and cloud-based attendance storage for individual student identification.*

Keywords—*Computer Vision, Automated Attendance, Object Detection, Real-Time Monitoring, Smart Classroom*